

Notes on HAR Categories:

<i>For 1 - 2 sites</i>	Single Hazard	Multi-hazard
Hazard Data is/are AVAILABLE	Simple	Simple
Hazard Data is/are NOT AVAILABLE	Complex	Complex
<i>For 3-4 sites</i>	Single Hazard	Multi-hazard
Hazard Data is/are AVAILABLE	Complex	Complex
Hazard Data is/are NOT AVAILABLE	Highly Technical	Highly Technical

**All requests consisting of 5 or more sites will be handled by GeMMs.*

Simple Transaction – pertains to one to two site requests for single or multiple hazards where assessment may proceed based on currently available and acceptable hazards information. (Processing Time ~3 working days)

OOD = 7 hrs/ within the day of receipt

VOD = 7 hrs/ within the day of receipt

Approver = 1 day

Complex Transaction – pertains to three to four site requests for single hazard or multiple hazards, provided that hazard data or information necessary for assessment are readily available. It can also be one to two site requests for multiple hazards provided one or two hazards require further analysis. (Processing Time ~7 working days)

OOD = 3 days

VOD = 2 days, 8 hrs

Approver = 1 day

Highly Technical Transaction – pertains to three or more site requests for multiple hazards, with two or more hazards that require detailed analysis. (Processing Time ~20 working days)

OOD = 12 days

VOD = 6 days

Approver = 1 day

EARTHQUAKE HAZARD ASSESSMENT

LOT DESCRIPTION; PROJECT NAME, LOCATION	GROUND RUPTURE	FISSURE <i>(if property is within ~1km of a mapped fissure)</i>	LIQUEFACTION	EARTHQUAKE- INDUCED LANDSLIDE	TSUNAMI
Survey Lot #; PROJECT NAME, Street, Barangay, City, Province	Safe; Approximately ___ kilometers ___ of the ___ Fault ** Write Fault Name as: <u>Fault Name:</u> <u>Segment Name</u> or <u>Fault Name</u> (only if same as Segment Name)	Safe; Approximately ___ meters ___ of a mapped fissure	Safe	Safe Note: If site is not generally flat; please provide elevation profile via Google Earth (If slope is less than 15 degrees = safe) For depositional zone: SAFE if horizontal distance is more than 3X of its elevation	Safe
	<i>(if transected by AF)</i> Prone; Transected by the ___ Fault <i>*(write if only a portion of the property is transected) at</i> the ___ portion of the property (western, eastern, northern, southern) <i>(if <= 5 meters)</i> Prone; Within the buffer zone; Approximately N meter/s (or less) ___ of the ___ Fault	<i>(if transected by a fissure)</i> Prone; Transected by a mapped fissure at the ___ portion of the property	<i>(for general susceptibilit y or MELP LIQN)</i> Susceptible Safe	<i>(for regional EIL)</i> Susceptible* Safe	<i>(GIA/READY / TsuHaMEI)</i> Prone; Within the tsunami inundation zone <i>*Kahit may partly safe</i>

	<i>(if property is near a fault with fissure)</i> Safe; Approximately ___ kilometers ___ of the ___ Fault	<i>(if property is near a fault with fissure)</i> Safe; Approximately ___ meters ___ of a mapped fissure that coincides the West Valley Fault	Largely Safe, Partly Susceptible Partly Safe, Largely Susceptible Partly Safe, Partly Susceptible	(regional EIL, for barangays/ large areas) Largely Safe, Partly Susceptible Partly Safe, Largely Susceptible Partly Safe, Partly Susceptible	<i>(coastal inner islands PH, low elevation no data available)</i> Low-lying coastal area is prone to local tsunami due to offshore fault or submarine landslide
	<i>(if centered between 2 traces of the same Fault name)</i> Safe; Approximately ___ meters ___ of the nearest segment of the ___ Fault *and approximately ___ meters ___ of another segment <i>*Note worth mentioning only if both segments are within -meter distance only</i>	<i>(if within 5 meters/ buffer zone)</i> Prone; Within the buffer zone; Approximately N meter/s (or less) ___ of a mapped fissure	<i>(for READY LIQN)</i> Least Susceptible Moderately Susceptible Highly Susceptible Largely Least Susceptible/ Moderately Susceptible/ Highly Susceptible, Partly Safe or Largely Safe, Partly Least Susceptible/ Moderately Susceptible/	<i>(for READY and Deterministic EIL projects)</i> Least Susceptible Moderately Susceptible Highly Susceptible	<i>(coastal E-W PH, low elevation no data available)</i> Low-lying coastal area is within the tsunami inundation zone

			Highly Susceptible Note: "Largely" = >70% coverage, anything less is "Partly"		
	<i>(if nearest fault is unnamed)</i> Safe; Approximately ___ kilometers ___ of an unnamed fault traversing ___ (Brgy/Muni) ___ (Muni/Prov) ___ Ex. Safe; Approximately ___ kilometers ___ of an unnamed fault traversing Mauban, Sampaloc, Tayabas and Pagbilao, Quezon Or Prone; Transected by an unnamed fault traversing Antipolo City, Baras and Tanay, Rizal*. at the ___ of the property		<i>(for GMMA-READY LIQN)</i> Low Potential Moderate Potential High Potential Note: Regardless of coverage, we do not say "partly/largely"	Largely Least Susceptible/ Moderately Susceptible/ Highly Susceptible, Partly Safe or Largely Safe, Partly Least Susceptible/ Moderately Susceptible/ Highly Susceptible Note: "Largely" = >70% coverage anything less is "Partly"	<i>(for large areas/brgy/municipal assessment)</i> Low-lying coastal area is within the tsunami inundation zone

	<i>(if nearest fault is offshore or on adjacent island)</i> Safe; Approximately ___ kilometers ___ of <u>nearest offshore fault</u> and approximately ___ kilometers ___ of <u>"nearest Fault onland/ within the same land as site; otherwise just assess nearest offshore)"</u>		<i>(if site is completely covered with multiple hazard zones)</i> Least to Moderately Susceptible/ Moderate to Highly Susceptible/ Least to Highly Susceptible	<i>(if site is completely covered with multiple hazard zones)</i> Least to Moderately Susceptible/ Moderately to Highly Susceptible/ Least to Highly Susceptible*	
	<i>(if nearest fault is offshore and unnamed fault)</i> Safe; Approximately ___ kilometers ___ of an unnamed offshore fault in <u>Location</u> and approximately ___ kilometers ___ of <u>"nearest Fault onland"</u>		<i>(for large areas/brgy/ municipal assessment)</i> Low-lying area is susceptible to liquefaction	<i>(*if within depositional zone)</i> Prone; Within the depositional zone *; Within the depositional zone	
				<i>(for large areas/brgy/ municipal assessment)</i> Steep and mountainous area is susceptible to earthquake-induced landslides	

VOLCANO HAZARD ASSESSMENT

LOT DESCRIPTION; PROJECT NAME LOCATION	NEAREST ACTIVE VOLCANO	LAHAR	PYROCLASTIC FLOW	LAVA FLOW
	Approximately __ kilometers of _____ Volcano	Safe	Safe	Safe
	Approximately __ kilometers of Kanlaon Volcano	(KANLAON) Highly Prone/ Least Prone/ Moderately Prone	(KANLAON) Prone/ Prone; Within buffer zone	(KANLAON) Highly Prone/ Least Prone
	Approximately __ kilometers of Mayon Volcano	(MAYON) Highly/ Moderately/ Least Prone	(MAYON) Prone	(MAYON) Prone
	Approximately __ kilometers of Pinatubo Volcano	(PINATUBO) Zone 1/ Zone 2/ Zone 3/ Zone 4/ Zone 5	(PINATUBO) Prone	
	Approximately __ kilometers of Banahaw Volcano	(BANAHAW) Highly/ Moderately/ Least Susceptible	(BANAHAW) Susceptible	
	Approximately __ kilometers of Bulusan Volcano	(BULUSAN) Susceptible/ Highly/ Moderately/ Least Susceptible	(BULUSAN) Highly/ Moderately to Least Susceptible	(BULUSAN) Highly Susceptible
	Approximately __ kilometers of Cabalian Volcano	(CABALIAN) Safe or Highly Prone/Moderately Prone	(CABALIAN) Safe or Prone	

(HIBOK-HIBOK VOLCANO ONLY)

LOT DESCRIPTION; PROJECT NAME LOCATION	NEAREST ACTIVE VOLCANO	PYROCLASTIC FLOWS AND LATERAL BLASTS	LAVA FLOW	LAHARS AND FLOODS	AIRFALL TEPHRA AND BALLISTIC PROJECTILES
	Approximately __ kilometers __ of Hibok-Hibok Volcano	Safe or Prone	Safe or Prone	Safe or Prone	Safe or Prone

(TAAL VOLCANO ONLY)

LOT DESCRIPTION; PROJECT NAME LOCATION	NEAREST ACTIVE VOLCANO	FISSURE (for Areas near Taal only) AGONCILLO, LEMERY, TAAL, TALISAY, TANAUAN, SAN NICOLAS	BASE SURGES	BALLISTIC PROJECTILES	VOLCANIC TSUNAMI
	Approximately __ kilometers __ of Taal Volcano	Approximately __ kilo/meters __ of a mapped fissure	Safe or Prone	Safe or Prone	Safe or Prone
		Prone; Transected at the __ portion by a mapped fissure			
			Prone; Within the base surge buffer zone		

(IRIGA VOLCANO ONLY)

LOT DESCRIPTION; PROJECT NAME LOCATION	NEAREST ACTIVE VOLCANO	LAHAR	PYROCLASTIC FLOW	LAVA FLOW	VOLCANIC TSUNAMI
	Approximately __ kilometers __ of Iriga Volcano	Safe or Prone	Safe or Prone	Safe or Prone	Safe or Prone

	High Danger Zone/ Moderate Danger Zone/ Low Danger Zone / within the Buffer Zone; Approximately ___ kilometers ___ of Iriga Volcano				
--	--	--	--	--	--

****Notes:**

1. For cases when the assessment does not appear in the columns, please check the number of characters permitted for the line within the column. You might have a very long word that doesn't fit in the specified columns. Kindly edit and put "`-<br>`" to truncate the long word (where appropriate), this will put the succeeding characters into the next line, and then write the succeeding parts of the words/assessment accordingly.

(Ex. *Approximately* → *Approx*-`<br>`*mately* =

Approx- mately

2. For HARs with more than 2 pages, please adjust paper size accordingly to ensure that the contents are properly placed.

In hasadmin, Manage Request > Input Request number > Click the pencil/edit icon
> Select paper size > Click Save Button



Paper Size

A4

✓ Legal

3. For transected HAMs to be released to the client, you may use the provided map document file for the layout found at:

F:\TECTONICS\ARCGIS\PROJECT_FILES

[\(.mxd\)\TEMPLATES\2021_OHAS\ManualAssessmentTemp_forOHAS.mxd](F:\TECTONICS\ARCGIS\PROJECT_FILES\TEMPLATES\2021_OHAS\ManualAssessmentTemp_forOHAS.mxd) (ArcmapV10.5) or you may download the latest mxd in the GDrive

<https://drive.google.com/drive/folders/1p6xOs8AYON-896gsSjjmx6fYO3iTKhiV>

Please make sure to use the appropriate basemap and zoom in up to (at most) 1:1K scale only (for WVF) and 1:5K for other traces.

4. For cases where the site/ some sites is/are **nearer** a POTENTIALLY ACTIVE VOLCANO: See VOL HAR Template

Conditions:

1. If both AV and PAV are within 50km but PAV is nearer, assess both AV and PAV; consult VOOD.
2. If both AV and PAV are within 50km but AV is nearer, assess AV only.
3. If AV and PAV are more than 50km away from the site, assess AV only.

EXPLANATION AND RECOMMENDATION [Standard]

EARTHQUAKE HARS

1 1 1 1 1 1	<i>(for assessment based on VM)</i> All hazard assessments are based on the latest available hazard maps and on the location indicated in the vicinity map provided. <i>(if coordinates/kmz provided were used for assessment)</i> All hazard assessments are based on the latest available hazard maps and on the location indicated by the coordinates provided. <i>(if coordinates/kmz and VM were provided and used for assessment)</i> All hazard assessments are based on the latest available hazard maps and on the location indicated in the vicinity map and by the coordinates provided. <i>(for barangay/municipal assessment)</i> Barangay hazard assessment is made when an appropriate vicinity map is not provided. Please note that barangays are located using the PSA (2016/2020). All hazard assessments are based on the latest available hazard maps. Municipal hazard assessment is made when an appropriate vicinity map is not provided. Please note that municipalities are located using the PSA (2016/2020). All hazard assessments are based on the latest available hazard maps. <i>(no VM provided)</i> All hazard assessments are based on the latest available hazard maps. In the absence of vicinity maps, the location of the property used were those indicated in Google Earth, Google Maps, or Wikimapia.
2a 2b	<u>Ground rupture hazard</u> assessment is the distance to the nearest known active fault. The recommended buffer zone, or Zone of Avoidance, against <u>ground rupture hazard</u> is at least 5 meters on both sides of the active fault or from its zone of deformation. <i>(add if with FISSURE assessment ONLY)</i> Fissure, with its related ground subsidence, may not be a potential source of earthquakes. However, fissures and ground subsidence may develop further if the condition that generated them continues to persists, or as a response to strong ground shaking. The recommended buffer zone, or Zone of Avoidance, against fissuring is at least 5 meters on both sides of the fissure.
3 3	All sites may be affected by strong <u>ground shaking</u>. <i>(For PALAWAN and SULU ONLY)</i> All sites may be affected by <u>ground shaking</u>.
4 4	<u>Ground shaking hazard</u> can be mitigated by following the provisions of the National Building Code and the Structural Code of the Philippines. <i>(if susceptible to LIQN hazard)</i> <u>Ground shaking and liquefaction hazards</u> can be mitigated by following the provisions of the National Building Code and the Structural Code of the Philippines.
5	<i>(if susceptible to EIL hazard/within depositional zone)</i>

	Avoidance is recommended for sites with <u>earthquake-induced landslide hazard</u> unless appropriate engineering interventions are in place.
5,6	<i>(if prone to TSU hazard)</i> Tsunami threat to people's lives can be addressed by community preparedness and tsunami evacuation plan. Advice for tsunami evacuation comes from public agencies and local governments. But more importantly, coastal communities must learn to evacuate themselves when they recognize the three natural signs of tsunami, namely 1) strong ground shaking, 2) unusual rise or fall of sea level, and 3) strong or unusual sound coming from the sea.
5,6,7	<i>(single site assessment)</i> This hazard assessment supersedes any previous assessment made by this office regarding the site.
5,6,7	<i>(multiple sites assessment)</i> This hazard assessment supersedes any previous assessment made by this office regarding the sites
	<i>(for previously assessed, different assessment)</i> This hazard assessment supersedes previous assessment, HAS-MM-YY-xxx dated DAY-MM-YY, made by this office regarding the site.

VOLCANO HARs (AV only)

1	<i>(for assessment based on VM)</i> All hazard assessments are based on the latest available hazard maps and on the location indicated in the vicinity map provided.
1	<i>(if coordinates/kmz provided were used for assessment)</i> All hazard assessments are based on the latest available hazard maps and on the location indicated by the coordinates provided.
1	<i>(if coordinates/kmz and VM were provided and used for assessment)</i> All hazard assessments are based on the latest available hazard maps and on the location indicated in the vicinity map and by the coordinates provided.
1	<i>(for barangay/municipal assessment)</i> Barangay hazard assessment is made when an appropriate vicinity map is not provided. Please note that barangays are located using the PSA (2016/2020). All hazard assessments are based on the latest available hazard maps.
1	Municipal hazard assessment is made when an appropriate vicinity map is not provided. Please note that municipalities are located using the PSA (2016/2020). All hazard assessments are based on the latest available hazard maps.
1	<i>(no VM provided)</i> All hazard assessments are based on the latest available hazard maps. In the absence of vicinity maps, the location of the property used were those indicated in Google Earth, Google Maps, or Wikimapia.
2	<i>(single site assessment)</i> {AV} Volcano is the nearest identified active volcano to the site.

2	<i>(multiple site assessment)</i> {AV} Volcano is the nearest identified active volcano to the sites.
3a 3a 3^ 3b- 3c 3*	<i>(if less than <50km/within watershed, with PDZ: Taal, Mayon, Bulusan, Kanlaon, Hibok-hibok)</i> The site is <u>outside</u> the ___-kilometer Permanent Danger Zone of _____ volcano. The site is <u>inside</u> the ___-kilometer Permanent Danger Zone of _____ volcano. <i>(for Pinatubo only)</i> The site is <u>inside</u> the 10-kilometer danger zone of Pinatubo volcano. The site is <u>outside</u> the 10-kilometer danger zone of Pinatubo volcano. <i>(if PDZ is mentioned)</i> The Permanent Danger Zone (PDZ) is a zone that can always be affected by small-scale eruptions of _____ Volcano. Human settlement is not recommended within the PDZ. <i>(ADD if inside TAAL PDZ)</i> Taal Volcano Island is designated as the Permanent Danger Zone of the volcano. <i>(for AV with no available Hazard Map/no P-DZ but within ~10 km radius of the volcano -provided not near a river channel)</i> The site is within the delineated 10-kilometer radius of the volcano and is considered as highly prone to eruptive processes and products in the event of an eruption.
3,4 3,4	<i>(for AV with no P-DZ; site is greater than >50km or outside watershed)</i> Considering the distance of the site/s from the volcano, the site/s is/are safe from volcanic hazards such as pyroclastic density currents, lava flows, and ballistic projectiles that may originate from the volcano. <i>(Surrounding areas outside the Volcano/ AV Island (nearest BILIRAN & HIBOK-HIBOK, etc)</i> Considering the distance of the site from the volcano, the site is safe from erupted products of a hazardous volcanic eruption.
4/5a 4/5b 4/5c	<i>(For near TAAL areas with Fissure Assessment Only: AGONCILLO, LEMERY, TAAL, TALISAY, TANAUAN, SAN NICOLAS)</i> Volcanic fissures identified are based on results from detailed field investigations. Hazard assessment may change as DOST-PHIVOLCS scientists gather more information on fissure occurrences. Fissures may develop further if the condition that generated them persists, as a response to geological processes. For land-use planning and disaster risk reduction purposes, at least 5 meters on both sides of the fissure or at the edge of the deformation zone may be considered as a Zone of Avoidance. Stakeholders are advised to report to the Institute fissures that may be observed at Taal Volcano and vicinity that need to be reflected in the fissure map.
4,5,6	<i>(if with hazard)</i> LAVA FLOWS Lava flows are rivers of hot molten rock that can burn and bury land and make these lands unusable for a long period of time. PDCs/BASE SURGES

4,5,6, 7	Pyroclastic Density Currents (PDCs), such as flows, surges and base surges are turbulent, fast-moving hot mixtures of volcanic fragments and gas that sweep down/away from an erupting volcano. PDCs can cause loss of life due to burning, impact force, abrasion, asphyxiation, and extreme physical injuries, burial, and damage to agriculture, livestock, and properties.
4b,5b, 6b,7b	(ADD clause if prone to PDCs and/or lava flows) Avoidance is recommended for site/sites that may potentially be affected by pyroclastic density currents (PDCs) and lava flows.
4,5	BALLISTIC PROJECTILES Ballistic projectiles are fragments thrown at a close distance from the vent/crater of an erupting volcano. Ballistic projectiles can cause loss of life, physical injuries, and damage to properties due to forceful impact when hitting ground.
4,5,6	VOLCANIC TSUNAMI Volcanic tsunami can be generated by the sudden displacement of sea/lake water by the entry of large masses of volcanic material, underwater explosion, and/or rapid ground deformation during volcanic activity. Volcanic tsunamis can cause inundation and washout of communities, drowning, physical injuries, and damage to infrastructure and properties.
4b,5b, 6b	(ADD clause if prone to volcanic tsunami) Tsunami threat related to the volcanic unrests can be addressed by community preparedness and tsunami evacuation plan, with special focus on shoreline communities.
4,5,6	LAHARS (GENERIC) Lahars are rapidly flowing mixtures of volcanic sediment and water in rivers that drain down volcanoes, typically produced by prolonged and intense rainfall. Lahars can cause extensive burial or washout of communities that can lead to physical injuries and loss of life, damage to built environment, agriculture, livestock, infrastructure, properties, as well as long-term flooding and siltation in river systems.
4b,5b, 6b	(ADD clause if prone to lahars) Lahar threat to people's lives can be addressed by 1) observing or implementing legal easement adjacent to river banks, as provided in existing laws, ordinances and land-use plans, and 2) community preparedness and evacuation plan. At-risk communities must learn to evacuate themselves when lahar threats are imminent.
4	PINATUBO (for lahar hazard only) The lahar hazards assessment for Pinatubo Volcano is a long-term lahar hazard projection based on the volcano's morphology, eruption history, as well as its tendency for rapid morphologic changes during periods of intense and/or prolonged rainfall, or after a major lahar event.
5	<u>Zone 1</u> includes areas at high susceptibility to large-magnitude lahars (more than 10 million cubic meters per event) and sediment-laden (muddy) stream flows and flashfloods.
5	<u>Zone 2</u> includes areas at low susceptibility to large-magnitude lahars, moderate to high susceptibility to small-magnitude lahars, and high susceptibility to sediment-laden (muddy) stream flows and/or flashfloods.

5	<u>Zone 3</u> includes areas at low susceptibility to small-magnitude lahars and high susceptibility to sediment-laden (muddy) stream flows.
5	<u>Zone 4</u> are areas safe from lahars but susceptible to sediment-laden (muddy) stream flows or siltation.
5	<u>Zone 5</u> are areas safe from lahars but susceptible to persistent (more than one week) or recurrent flooding and/or backflooding due to drainage blockage.
6	Lahar threat to people's lives can be addressed by 1) observing or implementing legal easement adjacent to river banks, as provided in existing laws, ordinances and land-use plans, and 2) community preparedness and evacuation plan. At-risk communities must learn to evacuate themselves when lahar threats are imminent.
4	MAYON (if with hazard only) The lahar hazards assessment for Mayon Volcano is a long-term lahar hazard projection based on the volcano's morphology, eruption history, as well as its tendency for rapid morphologic changes during periods of intense and/or prolonged rainfall, or after a major lahar event.
5	Areas <u>highly prone to lahars</u> are located adjacent to an active river channel that may be a potential lahar avenue in the event of an explosive eruption and additional lahar source materials become available.
5	Areas <u>moderately prone to lahars</u> are located in medial or distal portions of lahar fans, with a moderate probability of being affected by lahars and a high probability of being affected by sediment-laden streamflows.
5	Areas <u>least prone to lahars</u> are zones located in distal portions of lahar fans, or between river channels, with a low probability of being affected by lahars and by sediment-laden stream flows.
6	Lahar threat to people's lives can be addressed by 1) observing or implementing legal easement adjacent to river banks, as provided in existing laws, ordinances and land-use plans, and 2) community preparedness and evacuation plan. At-risk communities must learn to evacuate themselves when lahar threats are imminent.
4,6,7	In case of future eruptions of {AV} Volcano and other nearby volcanoes, the site/s may be affected by tephra fall/ ashfall depending on the height of the eruption plume and prevailing wind direction at the time of eruption. Generally, tephra fall/ ashfall is thicker near the eruption center and thins away from the volcano. Ash can cause widespread infrastructural, agricultural, and property damages, and negative health impacts.
5,8	<i>(single site assessment)</i> This hazard assessment supersedes any previous assessment made by this office regarding the site.
5,8	<i>(multiple sites assessment)</i> This hazard assessment supersedes any previous assessment made by this office regarding the sites
5,8	<i>(for previously assessed, different assessment)</i> This hazard assessment supersedes any previous assessment, HAS-MM-YY-xxx dated DAY-MM-YY, made by this office regarding the site.

VOLCANO HARs (with nearest PAV < 50kms)

Project Name/Lot Description	Volcano Distance	Lava Flow Hazard	Pyroclastic Density Current (PDC) Hazard	Lahar Hazard
(AV/PAV Distance only if no hazard map available)	Approximately X kilometers ____ of AV			
	Approximately X kilometers ____ of PAV			
(AV/PAV with manual assessment/ or hazard maps available)	Approximately X kilometers ____ of AV	Prone/ Safe	Prone/ Safe	Prone/ Safe
	Approximately X kilometers ____ of PAV	Prone/ Safe	Prone/ Safe	Prone/ Safe
(AV/PAV Assessment if AV with hazard maps, PAV no map available)	Approximately X kilometers ____ of AV	Prone/ Safe	Prone/ Safe	Prone/ Safe
	Approximately X kilometers ____ of PAV			

1	<i>(for assessment based on VM)</i> All hazard assessments are based on the latest available hazard maps and on the location indicated in the vicinity map provided. <i>(if coordinates/kmz provided were used for assessment)</i> 1 All hazard assessments are based on the latest available hazard maps and on the location indicated by the coordinates provided. <i>(if coordinates/kmz and VM were provided and used for assessment)</i> 1 All hazard assessments are based on the latest available hazard maps and on the location indicated in the vicinity map and by the coordinates provided. <i>(for barangay/municipal assessment)</i> 1 Barangay hazard assessment is made when an appropriate vicinity map is not provided. Please note that barangays are located using the PSA (2016/2020). All hazard assessments are based on the latest available hazard maps. 1 Municipal hazard assessment is made when an appropriate vicinity map is not provided. Please note that municipalities are located using the PSA (2016/2020). All hazard assessments are based on the latest available hazard maps. <i>(no VM provided)</i> 1 All hazard assessments are based on the latest available hazard maps. In the absence of vicinity maps, the location of the property used were those indicated in Google Earth, Google Maps, or Wikimapia.
2	<i>(single site assessment)</i> {AV} Volcano is the nearest identified active volcano to the site.
2	<i>(multiple sites assessment)</i> {AV} Volcano is the nearest identified active volcano to the sites.

3a 3b 3c	<i>(if with PDZ: Taal, Mayon, Bulusan, Kanlaon, Hibok-hibok)</i> The site is <u>outside</u> the ____-kilometer Permanent Danger Zone of _____ volcano. The site is <u>outside</u> the 10-kilometer danger zone of Pinatubo volcano. <i>(Pinatubo only)</i> <i>(for AV with no P-DZ; site is outside watershed)</i> Considering the distance of the site/s from the volcano, the site/s is/are safe from volcanic hazards such as pyroclastic density currents, lava flows, and ballistic projectiles that may originate from the volcano. <i>(Surrounding areas outside the Volcano/ AV Island (ex. BILIRAN and HIBOK-HIBOK, etc)</i> Considering the distance of the site from the volcano, the site is safe from erupted products of a hazardous volcanic eruption.
4	In case of future eruptions of {AV} Volcano and other nearby volcanoes, the site/s may be affected by tephra fall/ ashfall depending on the height of the eruption plume and prevailing wind direction at the time of eruption. Generally, tephra fall/ ashfall is thicker near the eruption center and thins away from the volcano. Ash can cause widespread infrastructural, agricultural, and property damages, and negative health impacts.
5 5	<i>(PAV with hazard map)</i> {PAV} Volcano is currently classified by DOST-PHIVOLCS as a potentially active volcano, which is morphologically young-looking but with no historical or analytical records of eruption. <i>(If PAV has no hazard map)</i> {PAV} Volcano is currently classified by DOST-PHIVOLCS as a potentially active volcano, which is morphologically young-looking but with no historical or analytical records of eruption, and therefore, has yet to be determined eruptive and hazard potential its eruptive and hazard potential is yet to be determined.
6 6,7 7,8 6,7,8 6,7,8	<i>(if site is assessed as prone to...)</i> LAVA FLOWS Lava flows are rivers of hot molten rock that can damage built environments and can make lands unusable for a long period of time. PDCs/BASE SURGES Pyroclastic Density Currents (PDCs), such as flows, surges and base surges are turbulent, fast-moving hot mixtures of volcanic fragments and gas that sweep down/away from an erupting volcano. PDCs can cause loss of life, extreme physical injuries, burial, and damages to agriculture and properties. (ADD clause if prone to PDCs and/or lava flows) Avoidance is recommended for site/sites that may potentially be affected by PDCs and lava flows. BALLISTIC PROJECTILES Ballistic projectiles are large fragments thrown out at a distance from the vent/crater of an erupting volcano. Damages related to ballistic projectiles can lead to loss of life, physical injuries, and damage to properties. VOLCANIC TSUNAMI Volcanic tsunami can be generated by the sudden displacement of sea/lake water by the entry of large masses of volcanic material, and rapid ground deformation during volcanic

	activity. Volcanic tsunamis can cause inundation and washout of communities, drowning, physical injuries, and damages to infrastructure and properties
7,8,9	<u>(ADD clause if prone to volcanic tsunami)</u> Tsunami threat related to the volcanic unrests can be addressed by community preparedness and tsunami evacuation plan, with special focus on shoreline communities,
8,9	<u>LAHARS (GENERIC)</u> Lahars are rapidly flowing mixtures of volcanic sediments and water in rivers that drain down volcanoes, typically produced by prolonged and intense rainfall. Lahars can cause extensive burial or washout of communities, physical injuries that could lead to loss of life, damages to the natural environment, agriculture, infrastructure, properties, and long-term flooding and siltation in river systems.
9,10	Lahar threat to people's lives can be addressed by 1) observing or implementing legal easement adjacent to river banks, as provided in existing laws, ordinances and land-use plans, and 2) community preparedness and evacuation plan. At-risk communities must learn to evacuate themselves when lahar threats are imminent.
10,11	This hazard assessment supersedes any previous assessment made by this office regarding the site/s.
10,11	<i>(for previously assessed, different assessment)</i> This hazard assessment supersedes any previous assessment, HAS-MM-YY-xxx dated DAY-MM-YY, made by this office regarding the site.

VOLCANO HAZARD INFORMATION AVAILABLE:

Taal - with VI PDZ | Base Surge | Ballistic Projectile | Volcano Tsunami (Seiche)

Mayon - with 6km PDZ | Lahar | Lava | PF

Pinatubo - with 10km DZ | Lahar | *PF (*PF data available ao1991; consult VOOD)

Kanlaon - with 4km PDZ | *Lahar | Lava | PF (*new data ao 2024)

Bulusan - with PDZ | Lahar | Lava | PF

Banahaw - *no PDZ | Lahar | PF

Cabalian - *no PDZ | Lahar | PF

Iriga - with PDZ | PDC | VOLCTSU

Isarog - *no PDZ | Lava | PDC | Lahar

Matutum - Lahar | PDC

Parker - Lahar | PDC

Hibok-Hibok - with PDZ | Ballistic Projectiles {| Lahar | Lava | PF |} - ongoing

Biliran - *no PDZ | Lava | PDC | Lahar

Musuan - *no PDZ | Lava | Lahar

Leonard Kniaeff - *no PDZ | Lahar | Lava | PDC

Bud Dajo - *no PDZ | Lava | Lahar

Cagua - *no PDZ | Lava | PDC | Lahar

Makaturing - no hazard map (if critical, consult with vggrd)

Iraya - *no PDZ | PDC | Lahar | Lava

Mahagnao - *no PDZ | PF | Lahar -note: lahar ang nasa hazard map/ data available in hasadmin assessment is PF only. Lahar channels reflected on the hazard map is available in geodatabase; need to consult Volcano group if within the vicinity (include explanation if site is within hazard area)

Ragang - *no PDZ | Lava | PDC | Lahar

Babuyan Claro - *no PDZ | Lava | PDC | Lahar

Camiguin de Babuyanes - *no PDZ | Lava | PDC | Lahar

Smith - *no PDZ | Lava

Didicas - *no PDZ | Lava | PDC | Lahar



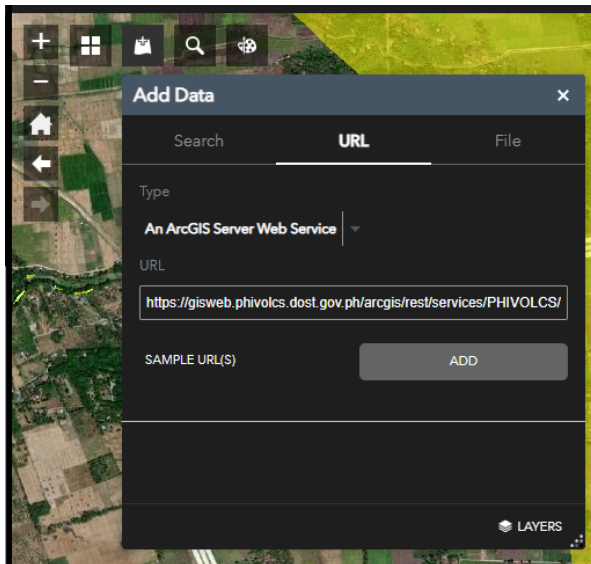
Danikko John Rivera
Jun 6, 2022



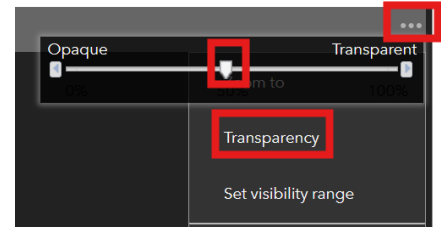
Volcano officer assigned per day. If the officer is not available, kindly inquire with Ms. Peej or OIC of the section that day for assistance. Thanks!

Monday- Ericson Bariso
Tuesday- Selwyn Cabaluna, Jr.
Wednesday- Valerie Shayne Olfindo
Thursday- Timothy John Daita
Friday- Marie Thess Quilalang-Gemal
Alternates:
Robjunelieaaa Lim
Raymond Patrick Maximo

PROVINCES WITH EIL DATA AVAILABLE



1. Click Add Data
2. Select Type: ArcGIS Server Web Service
3. Paste the WMS link in the URL box
4. Click ADD button
5. The map should appear on the Layer List
6. Adjust the transparency by clicking the ...



CAR (Abra; Apayao; Benguet; Ifugao; Kalinga; Mountain Province)

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_CAR/MapServer

Tarlac

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_Tarlac/MapServer

Panay and Guimaras

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_Panay_Guimaras/MapServer

Siquijor

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_Siquijor/MapServer

Pangasinan

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_Pangasinan/MapServer

Batangas

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_Batangas2024/MapServer

Cotabato

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_Cotabato2024/MapServer

Nueva Ecija

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_NuevaEcija2024/MapServer

La Union

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_LaUnion2024/MapServer

Biliran

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_Biliran2024/MapServer

Romblon

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_Romblon2024/MapServer

Occidental Mindoro

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_OccidentalMindoro2024/MapServer

Oriental Mindoro

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_OrientalMindoro2024/MapServer

Marinduque

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_Marinduque2024/MapServer

Albay

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_Albay/MapServer

Davao del Sur

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_DavaoDelSur2024/MapServer

Sorsogon

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_Sorsogon/MapServer

Samar Island

- https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_SamarIsland2025/MapServer

-----0-----0-----0-----0-----0-----0-----

PROVINCES WITH PRELIMINARY EIL DATA AVAILABLE IN *.Tiff/ WMS

(found in PIVS GISWeb Services or you may also copy the URL and load in OHAS Webapp for reference)

Preliminary EIL Hazard Data (for reference/checking)

- **Masbate (as of July 2023)**
https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EIL_Masbate_prelim2023/MapServer

For Manual Assessments you may use this project file found at:

**F:\TECTONICS\ARCGIS\PROJECT_FILES
 (.mxd)\TEMPLATES\2021_OHAS\ManualAssessmentTemp_forOHAS.mxd (ArcmapV10.5)**

For other HAS-related Files see:

<https://drive.google.com/drive/folders/1p6xOs8AYON-896gsSjjmx6fYO3ITKHiV>

WMS URLs (GISWEB) EARTHQUAKE-RELATED HAZARDS

Active Fault

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/ActiveFault/MapServer/>

Earthquake-Induced Landslide

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/EarthquakeInducedLandslide/MapServer/>

Regional EIL

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/regionalearthquakeinducedandslide/MapServer>

Ground Shaking

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/GroundShaking/MapServer/>

Liquefaction

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/Liquefaction/MapServer>

Tsunami

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/Tsunami/MapServer/>

VOLCANO-RELATED HAZARDS

Volcano Location (Active, Potentially Active, & Inactive Volcanoes)

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/VolcanoLocation/MapServer/>

Permanent Danger Zones (PDZ)

https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/DangerZone_permanent/MapServer

Pyroclastic Density Current

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/Pyroclastic/MapServer/>

Lahar

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/VolcanoLahar/MapServer/>

Lava

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCSPublic/Lava/MapServer/>

Base Surge

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/BaseSurge/MapServer/>

Ballistic Projectile

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/BallisticProjectile/MapServer/>

Volcano Tsunami/Seiches

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/Seiches/MapServer/>

Taal Volcano Fissures

OTHERS

MM Imagery (NAMRIA, 2011)

https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/NAMRIA/Manila_Imagery/ImageServer

Building Footprints (NAMRIA, 2011)

https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/NAMRIA/building_footprints_mm/MapServer

QC Parcels (Assessed as of 2011)

https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PHIVOLCS/QC_PARCELS/MapServer

PSA Boundary - Provincial (PSA, 2016)

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PSA/Provincial/MapServer>

PSA Boundary - Municipal (PSA, 2016)

<https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PSA/Municipal/MapServer>

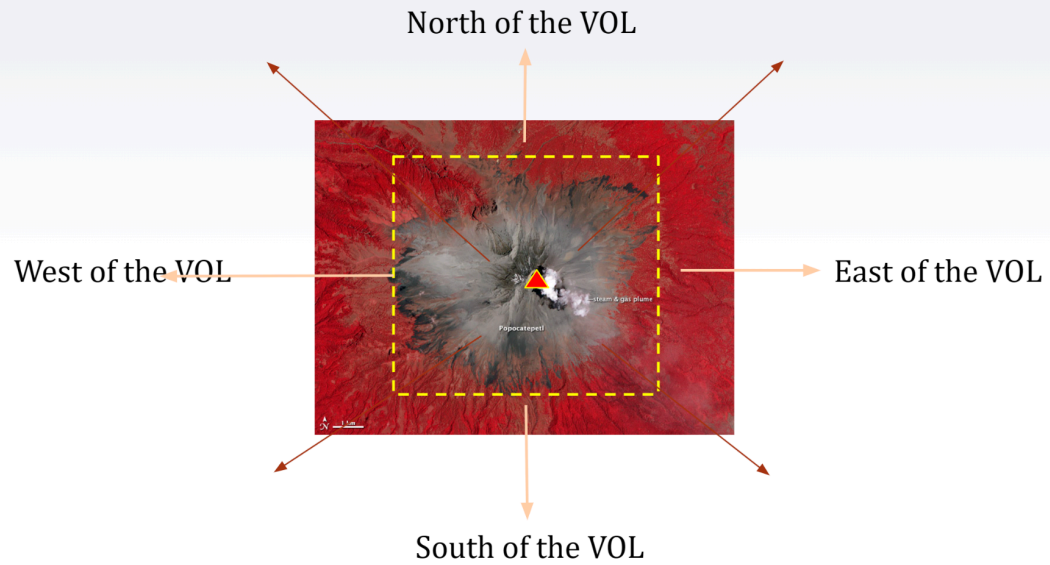
PSA Boundary - Barangay (PSA, 2016)

https://gisweb.phivolcs.dost.gov.ph/arcgis/rest/services/PSA/Barangay_ohas/FeatureServer

Notes on VOL Distance Direction

**Notes on VOL Distance Direction*

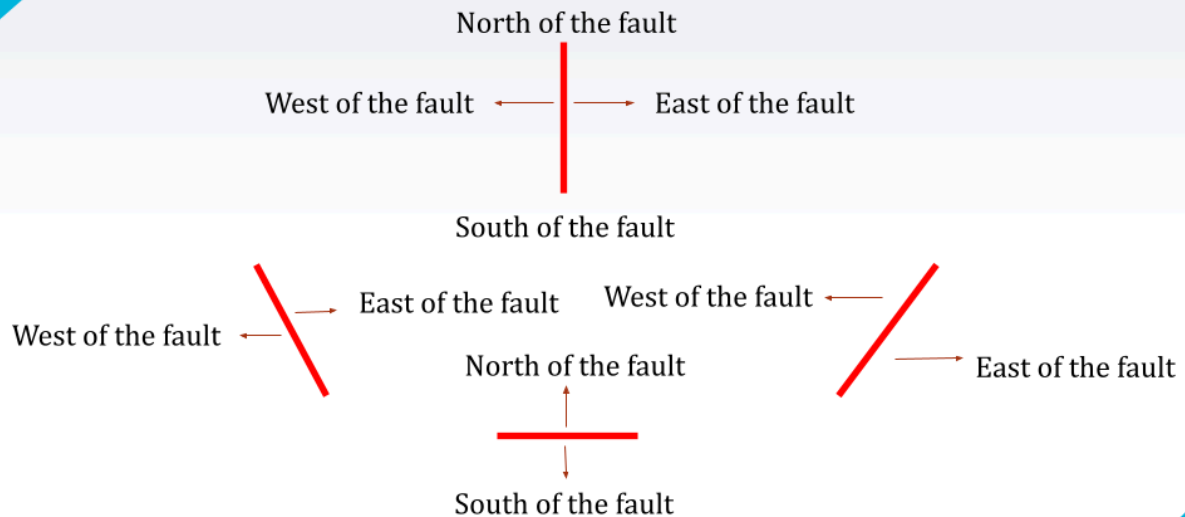
"The distance direction depends on the location of the site wrt the volcano edifice"



Notes on AF Distance Direction

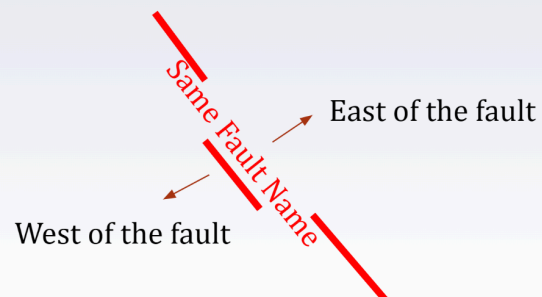
**Notes on AF Distance Direction*

"The distance direction depends on the overall fault orientation..."

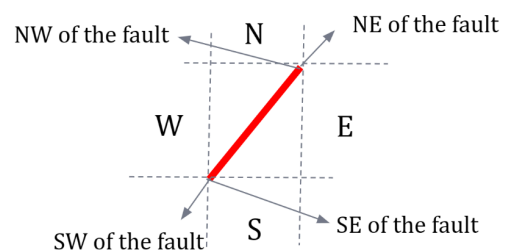
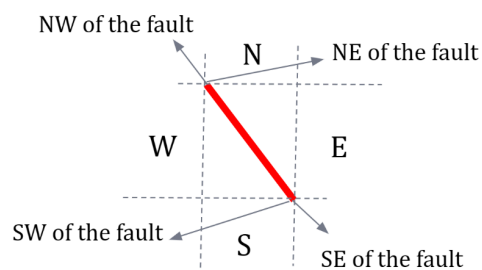


If the fault is segmented...

MEASURE BOTH:
 ~<1 km distance from
 2 different faults
 ~if in between segmented faults
*(standard wording to be used found in
 HAR_Template.docx)*



If site is beyond the edge of the fault...



PROVINCES WITH EARTHQUAKE HAZARD INFORMATION AVAILABLE

Provincial Maps	Earthquake Related Hazards				
	AFT	EIL	GSH	LIQ	TSU
Ilocos Norte	2021	2013	2013	2013	2023
Ilocos Sur	2021	2012	2012	2012	2023
La Union	2021	2024	2022	2018	2016
Pangasinan	2021	2021	2021	2025	2021
Batanes			2024	2020	2017
Cagayan	2021	2013	2013	2013	2007
Isabela	2021	2012	2012	2012	2007
Nueva Vizcaya	2019	2012	2012	2012	
Quirino	2019	2013	2013	2013	
Aurora	2019	2009	2024	2009	2009
Bataan			2024	2018	2017
Bulacan	2021	2014	2024	2024	2024
Nueva Ecija	2021	2024	2022	2018	
Pampanga		2010	2011	2010	2010
Tarlac	2021	2021	2021	2021	
Zambales	2021	2011	2011	2011	2023
Batangas	2023	2024	2022	2023	2017
Cavite	2021	2014	2014	2014	2023
Laguna	2021	2023	2023	2023	
Quezon	2021		2024	2019	2007
Rizal	2021	2014	2014	2014	
Marinduque	2018	2024	2022	2019	
Occidental Mindoro	2021	2024	2022	2018	2007
Oriental Mindoro	2021	2024	2022	2023	2007
Palawan				2018	2007
Romblon	2018	2024	2024	2020	
Albay	2018	2024	2024	2025	2021
Camarines Norte			2024	2018	2021
Camarines Sur	2021		2024	2018	2021
Catanduanes		2012	2012	2012	2017
Masbate	2018		2024	2020	
Sorsogon	2021		2024	2018	2021
Aklan	2018	2021		2018	2007
Antique	2021	2021	2012	2012	2012
Capiz	2019	2021		2018	
Guimaras		2021		2018	2007
Iloilo	2020	2021	2013	2013	2013
Negros Occidental	2021	2018	2022	2018	2018
Bohol	2021	2018	2017	2017	2017
Cebu	2021	2017	2017	2017	2017
Negros Oriental	2021	2017	2022	2017	2017
Siquijor		2021		2018	2007
Biliran		2024		2018	
Eastern Samar	2018	2010	2011	2010	2021
Leyte	2021	2007	2007	2007	2023
Northern Samar	2018	2010	2010	2010	2020
Samar	2018			2019	
Southern Leyte	2021	2007	2007	2007	2023

Provincial Maps	Earthquake Related Hazards				
	AFT	EIL	GSH	LIQ	TSU
Zamboanga Sibugay	2020			2021	2007
Zamboanga del Norte	2020			2021	2007
Zamboanga del Sur	2020			2021	2007
Isabela City					2007
Zamboanga City					2007
Bukidnon	2020		2024	2019	
Camiguin			2024	2021	2007
Misamis Occidental	2020		2024	2021	
Misamis Oriental	2020		2024	2023	
Lanao del Norte	2020		2024	2020	2007
Davao de Oro (Compostela Valley)	2021		2025	2021	
Davao del Norte	2021			2019	
Davao del Sur	2021	2024	2022	2024	2023
Davao Occidental	2021		ongoing	2019	2007
Davao Oriental	2018		2025	2019	2017
Davao City	2023	2019	2022	2018	2023
Cotabato (North Cotabato)	2020	2024	2022	2024	
Sarangani	2020		ongoing	2021	2007
South Cotabato	2020		ongoing	2019	2007
Sultan Kudarat	2020		ongoing	2021	2007
Cotabato City				2023	
General Santos City				2022	2007
Agusan del Norte	2021		ongoing	2021	
Agusan del Sur	2018		ongoing	2018	
Dinagat Islands		2008	2023	2008	2021
Surigao del Norte	2018	2008	2023	2008	2023
Surigao del Sur	2021	2008	2023	2008	2023
Abra	2020	2021	2013	2013	
Apayao	2019	2021		2018	
Benguet	2021	2021	2012	2012	
Ifugao	2019	2021		2020	
Kalinga	2019	2021		2020	
Mountain Province	2019	2021		2020	
Basilan				2018	2007
Lanao del Sur	2020			2021	2007
Maguindanao del Norte			2024		
Maguindanao del Sur	2020		2024	2021	2007
Sulu				2021	2007
Tawi -Tawi				2021	2007

NCR - National Capital Region	2015	2013	2014	2014	2014
-------------------------------	------	------	------	------	------

VOLCANOES with HAZARD MAPS/ INFORMATION AVAILABLE

	Active Volcanoes	Volcano Hazard Extent (PROVINCE/S)	Volcano Related Hazard Maps								
			VHM (Volcanic Hazard Map)	LAV (Lava Flow)	LHR (Lahar)	PDC (Pryroclastic Flow / Base Surges)	BAL (Ballistic Projectile)	VTS (Volcanic Tsunami)	VFI (Fissures)	TEP (Ashfall/ Tephra Fall)	PDZ (Permanent Danger Zone)
1	Babuyan Claro	Cagayan	2024 (PDC&LHR)	2024							
2	Banahaw	Laguna, Quezon			2023	2023					
3	Biliran (Anas)	Biliran	2023 (PDC&LHR)	2023							
4	Bud Dajo	Sulu									
5	Bulusan	Sorsogon		2016	2016	2016			2010	2016	
6	Cabalian	Southern Leyte			2023	2023					
7	Cagua	Cagayan	2023 (PDC/LHR)	2023							
8	Camiguin de Babuyan	Cagayan	2024 (PDC&LHR)	2024							
9	Didicas	Cagayan	2024 (PDC,LHR&LAV)								
10	Hibok-hibok	Camiguin		1988	1988	1988	2023				
11	Iraya	Batanes	2023 (PDC/LHR)	2023							
12	Iriga	Camarines Sur	2023 (PDC/VTS)								
13	Isarog	Camarines Sur		2023	2023	2023					
14	Kanlaon	Negros Oriental, Negros Occidental		2016	2016	2016					
15	Leonard Kniaseff	Davao del Norte, Compostella		2023	2023	2023					
16	Makaturing	Lanao del Sur									
17	Matutum	South Cotobato, Sarangani, Sultan Kudarat	2023 (PDC/LHR)								
18	Mayon	Albay		2018	2018	2018			2018		
19	Musuan (Calayo)	Bukidnon									
20	Parker	South Cotobato, Sarangani	2023 (PDC/LHR)								
21	Pinatubo	Bataan, Pampanga, Tarlac,			2022						2021
22	Ragang	Lanao del Sur, North Cotabato, Maguindanao		2024	2024	2024					
23	Smith	Cagayan		2024							
24	Taal	Batangas				2020	2020	2020	2023		

